Beyond Captioning: A Preference-Aligned Benchmark for Fine-Grained Reasoning and Spatiotemporal Geo-World Understanding

Ben Trovato*
G.K.M. Tobin*
trovato@corporation.com
webmaster@marysville-ohio.com
Institute for Clarity in Documentation
Dublin, Ohio, USA

Lars Thørväld
The Thørväld Group
Hekla, Iceland
larst@affiliation.org

Valerie Béranger
Inria Paris-Rocquencourt
Rocquencourt, France

Aparna Patel
Rajiv Gandhi University
Doimukh, Arunachal Pradesh, India

Huifen Chan
Tsinghua University
Haidian Qu, Beijing Shi, China

Abstract

Change analysis in spatiotemporal contexts is one of the core tasks in Earth observation. However, existing change captioning models mainly focus on describing shallow visual differences of the land surface, lacking a deep understanding of such issues as the driving forces behind changes and whether changes are semantically valid in real-world land-use scenarios. This paper proposes a task shift from change annotation to change interpretation and understanding, and introduces a new large-scale, highly informative benchmark dataset designed to achieve fine-grained reasoning and spatiotemporal world understanding in a multimodal setting. Our dataset goes beyond simple description and object detection, and further strengthens the deeply logical coupling between visual attributes and real-world land-use conditions: it models spatial, interactive, functional and foundational relations, and finely aligns high-level geographic semantics, real-world application scenarios and complex visual features. To improve reasoning reliability, we construct a process supervision mechanism based on positive–negative dual-track reasoning logic, which effectively mitigates the hallucination problem of multimodal models in complex geographic scenes. In addition, we build a large-scale negative sample set to reduce false positives in intra-class variation scenarios where no actual recognition is required. To address the inherent resolution limitations and classification ambiguities of satellite observation, we introduce an uncertainty-aware confidence score aligned with fine-grained human expert preferences, ensuring high consistency between model outputs and expert knowledge. Furthermore, we perform spatiotemporal future inference based on the above deep understanding and geographic expert knowledge, providing predictive support for geographic decision-making assistance. Extensive benchmark experiments show that our dataset not only provides a strong foundation for supervised fine-tuning (SFT), but also offers critical infrastructure for reasoning reward modeling and fine-grained error diagnosis in next-generation geographic world models. The dataset, generation code and baseline models are publicly available at: https://github.com/kaynqi/Beyond-Captioning.

CCS Concepts

• **Computing methodologies** → **Scene understanding**; *Computer vision tasks*; • **Information systems** → *Data sets*; • **Applied computing** → Earth and atmospheric sciences.

Keywords

multimodal reasoning, VLMs, spatiotemporal reasoning, semantic change interpretation, benchmark dataset, preference-aligned learning, process supervision, Earth observation

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1 Introduction

Earth observation and spatiotemporal change analysis form the foundation for understanding the dynamic evolution of the Earth, driving key applications such as urban planning, disaster response and ecological monitoring. In recent years, with the rapid advancement of multimodal large language models (MLLMs), remarkable progress has been made in remote sensing change captioning (RSCC). By converting visual differences between bi-temporal images into natural language, existing models have successfully bridged the gap between pixel-level changes and semantic descriptions.

Nevertheless, the fundamental bottleneck of the current change captioning paradigm lies in its superficiality. Existing datasets and models primarily concentrate on shallow visual discrepancies (e.g., "several buildings were constructed" or "vegetation disappeared"), lacking the deep logical capacity required to interpret the driving forces, functional transitions and semantic validity of changes in real land-use scenarios. They are inherently unable to meet real-world geographic demands. For instance, many complex tasks require not only filtering out pseudo-changes caused by seasonal

*Both authors contributed equally to this research.

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variations, illumination differences or sensor discrepancies, but also complying with practical application requirements—such as whether the changed areas satisfy constraints under professional standards and whether they fall within the scope of operational interest. Current models suffer from high false positive rates for non-target changes. In the context of satellite observation, due to inherent resolution constraints and classification ambiguity, purely descriptive models cannot deliver reliable, expert-level support for geographic decision-making. Furthermore, inherent limitations of satellite imagery, such as resolution restrictions and ambiguous seasonal intra-class variations, frequently lead to false alarms and high misjudgment rates, which are difficult to resolve within existing image captioning-based frameworks.

To bridge the gap between shallow visual description and high-level geographic semantic understanding, we advocate a transition from simplistic change captioning to finer-grained change interpretation and understanding. This paper presents Geo-BC, a novel large-scale and highly informative benchmark dataset. Focusing on multimodal land-use monitoring scenarios with broad scalable application demands, this dataset is constructed to evaluate and enhance preference-aware reasoning and spatiotemporal geographic world understanding in MLLMs for multi-level land-use modeling. Our benchmark goes beyond basic object detection and caption generation, deeply integrating complex visual attributes with real-world land-use semantics. It explicitly models spatial, interactive, functional and foundational relations, and elegantly combines high-level geographic semantics with intricate visual features.

To ensure logical rigor and reduce misjudgment, we pioneer a process supervision mechanism based on dual-track (positive and negative) reasoning logic. Additionally, we construct a large negative sample set specifically targeting pseudo-changes and intra-class variations, thereby substantially lowering false positives in non-essential scenarios. Considering the inherent ambiguity of remote sensing imagery, we introduce an uncertainty-aware confidence scoring system aligned with fine-grained human expert preferences, ensuring that model outputs conform to professional geographic knowledge. Building on this deep understanding, our dataset is further extended to spatiotemporal future extrapolation, providing predictive insights for downstream geographic decision-making.

In summary, our main contributions are as follows: The primary contributions of this work are:

(1) **A dataset for multimodal reasoning:** We introduce Geo-BC that reformulates bi-temporal remote-sensing change analysis as a multimedia reasoning task rather than a captioning task. Instead of rewarding surface-level descriptions, Geo-BC evaluates whether a model can perform grounded spatial, temporal, and causal reasoning over image pairs and natural-language queries.
(2) **A language-grounded annotation framework for structured change interpretation:** We introduce a structured six-step reasoning annotation framework that supports language-grounded spatial, temporal, and semantic interpretation of geographic change.
(3) **An explainable and preference-aware supervision pipeline:** We propose a decoupled multi-judge framework and step-level process scoring to supervise not only the final answer but also the intermediate reasoning trajectory, which improves explainability and enables preference-aligned training and diagnosis.
(4) **A practical benchmark for high-stakes multimedia applications:** By modeling semantically valid versus invalid changes, incorporating hard negatives, and introducing uncertainty-aware confidence scoring, the benchmark is designed for real-world multimedia applications in Earth observation.

2 Methodology

To move remote sensing change analysis beyond surface-level captioning, we construct a dataset generation and alignment pipeline that treats bi-temporal interpretation as a structured, judge-supervised reasoning process rather than a single free-form output. The core idea is to explicitly decompose geographic understanding into interpretable intermediate steps, evaluate those steps with specialized judges, and convert the resulting supervision into a preference-aligned dataset that supports both benchmark evaluation and downstream training.

To keep the main paper concise, we summarize only the key design choices and high-level statistics here. The supplementary material provides the complete task specification, annotation and judging protocols, manually verified quality-control evidence, extended dataset characterization, and additional examples necessary for full transparency and reproducibility.

2.1 Structured Multimodal Chain of Reasoning

A major limitation of conventional change captioning is that it maps a bi-temporal image pair directly to a final textual description, leaving the intermediate reasoning process implicit. This makes it difficult to determine whether a model has grounded its interpretation in visual evidence, reasoned consistently over spatial structure, distinguished meaningful change from nuisance variation, or calibrated its confidence under ambiguous observations. To address this limitation, we represent change understanding as a Structured Multimodal Chain of Reasoning.

Formally, each sample is represented by an input tuple

$$(P, S, Q, I), \tag{1}$$

where P denotes the system prompt, S the task instruction, Q the query, and I the bi-temporal image pair. Instead of predicting a final answer A in a single step, the policy model π_θ is required to generate a six-stage reasoning trajectory

$$R = \{r_1, r_2, r_3, r_4, r_5, r_6\}, \tag{2}$$

followed by the final output. This design makes the inference process explicit and creates the foundation for process-level evaluation, reward modeling, and fine-grained error diagnosis.

Step 1: Global perception (r_1). The first stage summarizes the overall scene and establishes a coarse semantic understanding of the bi-temporal input. This stage serves as a global contextual anchor before the model commits to localized or object-level interpretation.

Step 2: Instance-level visual grounding (r_2). The second stage identifies salient entities, regions, and attributes, and links them

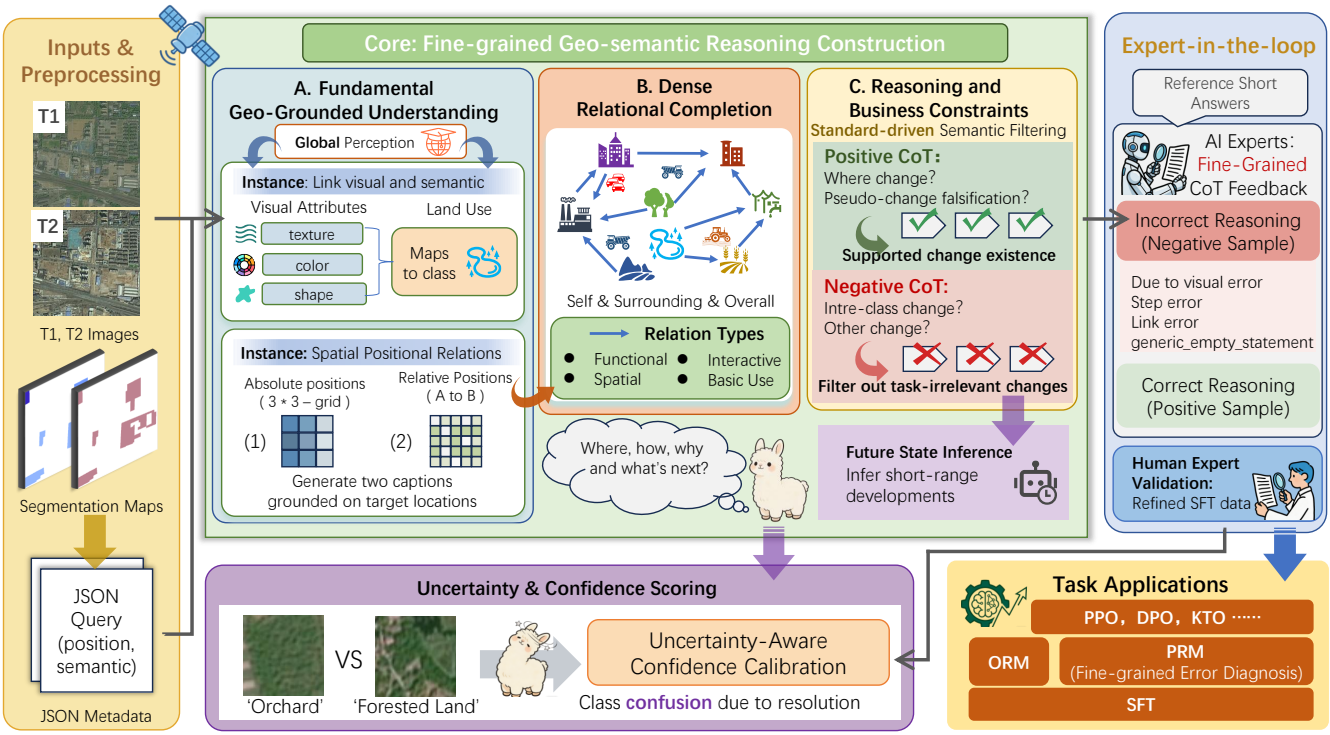

Figure 1: Overview of the proposed methodology. The illustration shows how bi-temporal remote-sensing inputs and auxiliary segmentation metadata are transformed into a structured reasoning process comprising global perception, instance grounding, dense relational completion, standards-driven semantic filtering, future-state inference, and uncertainty-aware confidence scoring, with expert-in-the-loop validation to support downstream training and benchmarking.

to concrete visual evidence in the image pair. When auxiliary geometric priors are available, grounding is further constrained by explicit region-level cues, which reduces unsupported references and improves spatial faithfulness.

Step 3: Spatial relation modeling (r_3). The third stage captures the relative positions, interactions, and topological organization of grounded entities. This stage converts local observations into a structured representation of scene layout and temporal evolution, which is essential for downstream geo-semantic reasoning.

Step 4: Causal and semantic reasoning (r_4). The fourth stage integrates visual evidence, grounded objects, and spatial structure to determine whether the observed differences correspond to a semantically meaningful change. This is the core reasoning stage in which the model moves from perception to interpretation.

Step 5: Future inference (r_5). The fifth stage extends the analysis from retrospective interpretation to forward-looking reasoning by predicting plausible short-term scene evolution conditioned on the current spatiotemporal state.

Step 6: Confidence correction (r_6). The final stage calibrates the output by explicitly accounting for uncertainty arising from limited resolution, class ambiguity, incomplete evidence, or other sources of epistemic uncertainty commonly encountered in remote sensing imagery.

2.2 Positive-Negative Dual-Track Reasoning

A central challenge in remote-sensing change interpretation is that visible difference is not equivalent to operationally valid land-use change. In real monitoring scenarios, the goal is not only to determine whether two images differ, but also to decide whether the observed difference should be counted as a target event under a predefined land-use taxonomy and application protocol.

We therefore introduce a positive-negative dual-track reasoning framework that separates evidence-supported change existence from standards-constrained change validity. The positive track establishes whether a candidate change hypothesis is grounded in visual, spatial, and temporal evidence. The negative track then evaluates whether the observed difference remains within the same task-relevant land-use class or subclass and should therefore be filtered out under the current monitoring objective. Importantly, intra-class variation is not assumed to be unimportant in a general geographic sense; rather, such variation is treated as outside the positive target definition of this benchmark when it does not cross the semantic boundary specified by the operational taxonomy. In our implementation, this taxonomy follows the national land-use classification standard GB/T 21010-2017. This design better reflects high-stakes monitoring settings in which the key challenge is not merely to notice any visual change, but to determine whether the

change constitutes a semantically valid and decision-relevant land-use transition. Consequently, the benchmark explicitly discourages models from mapping phenological variation, illumination effects, local texture shifts, or other within-class fluctuations to false functional transitions.

2.3 Decoupled Multi-Agent Criterion System

To construct a diverse yet stable candidate pool for each query, we use multiple multimodal foundation models(**Gemini 2.5 Pro, ChatGPT 4.1, LLaVA 7B, QVQ, and Qwen 2.5 VL 7B**), including high-quality proprietary systems and representative open-weight baselines, to generate candidate reasoning trajectories and final answers. All generator models are run with temperature set to 0 in order to maximize determinism, reduce unnecessary annotation variance, and improve reproducibility at dataset scale. This strategy allows us to combine complementary reasoning styles and error modes while avoiding over-reliance on a single generator family. The goal of multi-model generation is not to replace human expertise, but to provide a scalable first-pass candidate pool that can subsequently be filtered, ranked, and validated through structured judging.

We then evaluate these candidates using a *decoupled Mixture-of-Judges (MoJ)* rather than a single unified evaluator. Long multimodal reasoning traces must be judged along several dimensions simultaneously, including visual grounding, spatial consistency, change validity, uncertainty calibration, and standards compliance. We therefore distribute evaluation across three specialized agents, each responsible for a well-defined subset of the reasoning chain.

The step-to-judge mapping is defined as

$$\begin{aligned} \text{step}_1 &\leftarrow \text{VLM}, \\ \text{step}_2 &\leftarrow \text{VLM}, \\ \text{step}_3 &\leftarrow \text{LLM}, \\ \text{step}_4 &\leftarrow \text{LLM} + \text{Rule}_{\text{ontology}}, \\ \text{step}_5 &\leftarrow \text{LLM}, \\ \text{step}_6 &\leftarrow \text{Rule}_{\text{confidence}}. \end{aligned} \tag{3}$$

Judge 1: Visual-Language Expert. Judge 1 evaluates **Step 1** and **Step 2**, namely global perception and instance-level visual grounding. Its role is to verify that candidate trajectories are supported by the image pair and that referenced entities, regions, and attributes are grounded consistently in visual evidence.

Judge 2: Spatiotemporal Logic Expert. Judge 2 evaluates **Step 3**, **Step 4**, and **Step 5**, corresponding to spatial relation modeling, causal reasoning, and future inference. It focuses on the internal consistency of the spatial representation, the plausibility of the inferred causal chain, and the coherence of future-oriented reasoning.

Judge 3: Standards-Compliance Judge. Judge 3 evaluates the rule-sensitive parts of the trajectory, including ontology-sensitive reasoning in **Step 4**, uncertainty-aware calibration in **Step 6**, and the **final answer**. Its role is to verify that the candidate output remains consistent with geographic taxonomy, semantic constraints, and the required confidence behavior.

To assess annotation reliability, we randomly sampled 10% of the dataset and compared the AI-selected winning candidates with preference judgments provided by six human evaluators with remote-sensing research backgrounds. The human evaluation focused on whether the selected trajectories were better grounded, more semantically consistent, and more compliant with land-use standards than the rejected alternatives. We observed 92.0% agreement between the human preference judgments and the winners selected by the decoupled judging pipeline. This result suggests that the proposed multi-model annotation and multi-judge evaluation framework provides stable preference labels at scale while remaining largely consistent with expert human assessment.

2.4 Process Reward Modeling and Preference-Aligned Data Construction

Based on the decoupled MoJ system, we model the reward of the Structured Multimodal Chain of Reasoning at the *step level* and use it to construct a preference-aligned dataset. For each question and for each of the five generator models, the three judges produce structured evaluations over the relevant steps. These evaluations are then converted into scalar step scores and merged into a unified candidate record.

Let s_1 and s_2 denote the scores assigned by Judge 1, let s_3, s_4^{logic}, and s_5 denote the scores assigned by Judge 2, and let s_4^{rule} and s_6 denote the scores assigned by Judge 3. We define the process score as

$$s_{\text{proc}} = \frac{s_1 + s_2 + s_3 + s_4^{\text{logic}} + s_4^{\text{rule}} + s_5 + s_6}{7}. \tag{4}$$

This design reflects the fact that Step 4 sits at the intersection of open-ended reasoning and formal standards compliance, and therefore requires both logical and ontology-aware evaluation.

In addition to the process score, each candidate receives a hard answer score

$$s_{\text{ans}} \in \{0, 10\}, \tag{5}$$

where 10 denotes a correct final answer and 0 denotes an incorrect one. Candidates are then ranked in a two-stage manner: first by answer score, and second by process score. Formally, for the candidate set C_q associated with question q, we define

$$c_q^\star = \arg\max_{c \in C_q} \left(s_{\text{ans}}(c), s_{\text{proc}}(c) \right), \tag{6}$$

where the pair is compared in descending lexicographic order. This ranking strategy ensures that final-answer validity remains essential, while still rewarding trajectories that exhibit stronger perceptual grounding, better reasoning quality, and better compliance behavior.

The top-ranked candidate is treated as the supervised fine-tuning reference when it satisfies a minimum quality threshold. Specifically, we define the validity indicator

$$\mathbb{I}_{\text{valid}}(c_q^\star) = \mathbb{I}\left[s_{\text{ans}}(c_q^\star) = 10 \ \wedge \ s_{\text{proc}}(c_q^\star) > \tau \right], \tag{7}$$

where τ is the process-quality threshold. In our implementation, we set $\tau = 6.0$. If $\mathbb{I}_{\text{valid}}(c_q^\star) = 1$, the top-ranked candidate is retained as the *chosen* trajectory; otherwise, the question is marked as lacking a reliable chosen sample. All remaining candidates are retained as *rejected* trajectories. The resulting dataset preserves not only a single best answer, but also the full ranking structure over alternative trajectories, including detailed step scores and judge feedback.

This data format supports a broad range of downstream training objectives. The top-ranked chosen trajectory can be used for supervised fine-tuning (SFT); chosen/rejected pairs support preference

optimization methods such as DPO, PPO, GRPO, and KTO; and the step-level dense scores provide training signals for process reward modeling and fine-grained reasoning diagnosis. More broadly, because each sample contains multimodal context, a structured reasoning trajectory, stage-specific critiques, and a final preference ranking, the dataset can serve as infrastructure not only for benchmarking, but also for research on process supervision, alignment under formal constraints, and interpretable multimodal reasoning in Earth observation.

2.5 Dataset Characteristics and Statistical Analysis

The current benchmark contains 109,224 spatiotemporal image samples derived from GF-1, GF-2, GF-6, and GF-7 observations, forming a large-scale testbed for taxonomy-aware change interpretation. To support process-level supervision, each sample is associated with structured reasoning artifacts, yielding 327,672 inference trajectories with aligned scoring signals and 382,284 secondary evaluation statements generated by the decoupled judging framework. The overall dataset is deliberately skewed toward difficult non-target cases, with a positive-to-negative ratio of approximately 1:4, in order to reflect the realistic dominance of ambiguous or operationally irrelevant changes in practical monitoring scenarios. The present release covers 14 frequently used land-use categories under the national standard, with a planned expansion to the complete 25-category taxonomy. To further support high-end benchmarking, we additionally construct a manually curated Pro subset of 39,646 particularly challenging samples. This subset is designed for stress-testing fine-grained reasoning and semantic filtering, and therefore adopts a more balanced 1:1 positive-to-negative ratio.

The full dataset is divided into training, validation, and test sets in a 7:1:2 ratio. To avoid spatial leakage and inflated generalization performance, we enforce non-overlap in geographic locations across the three splits.

The dataset is constructed from multi-source Gaofen satellite imagery, including GF-1, GF-2, GF-6, and GF-7. These sensors provide complementary spatial resolutions and coverage scales: GF-1 offers 2 m / 8 m / 16 m imagery with 60 km × 60 km coverage; GF-2 provides 0.8 m / 3.2 m imagery with 45 km × 45 km coverage; GF-6 provides 2 m / 8 m imagery with 90 km × 90 km coverage; and GF-7 provides 0.65 m / 2.6 m imagery with 20 km × 20 km coverage. All sensors include panchromatic and multispectral observations, enabling fine-grained land-use change interpretation across heterogeneous spatial scales.

2.6 Experiments

Acknowledgments

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References

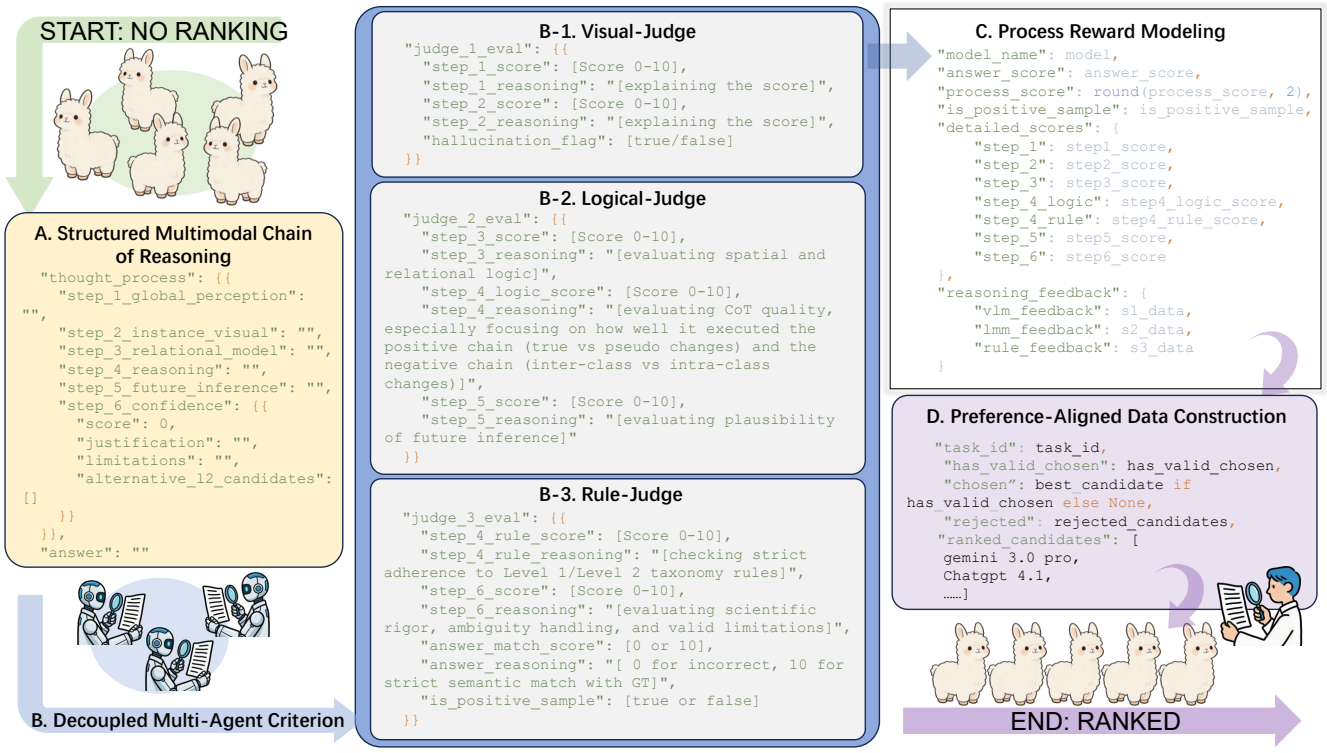

Figure 2: The illustration details the structured multimodal chain of reasoning, the decoupled three-judge evaluation modules for visual, logical, and rule-based assessment, the aggregation of stepwise scores into process rewards, and the final conversion of candidate outputs into ranked chosen/rejected samples for preference-aligned dataset construction.

Table 1: Unified comparison of six benchmarked MLLMs across the three-stage evaluation suite. Experiment I measures zero-shot basic change discrimination; Experiment II evaluates controlled zero-shot reasoning; Experiment III evaluates template-guided autonomous deep interpretation. Higher is better for all metrics except LHR.

Model	Experiment I			Experiment II				Experiment III					
	CDA ↑	CC-L1 ↑	CC-L2 ↑	VE ↑	LE ↑	CE ↑	LHR ↓	VP ↑	SL ↑	RA ↑	DD ↑	CC ↑	Overall ↑
LLaVA-OneVision	–	–	–	–	–	–	–	–	–	–	–	–	–
InternVL3-8B	–	–	–	–	–	–	–	–	–	–	–	–	–
Qwen2.5-VL-7B-Instruct	–	–	–	–	–	–	–	–	–	–	–	–	–
MiMo-V2-Flash	–	–	–	–	–	–	–	–	–	–	–	–	–
QVQ-Max	–	–	–	–	–	–	–	–	–	–	–	–	–
SkySense	–	–	–	–	–	–	–	–	–	–	–	–	–

Experiment I: CDA = Change Detection Accuracy; CC-L1/CC-L2 = Classification Consistency at Level-1/Level-2 categories.
Experiment II: VE = Visual Evidence; LE = Logical Evidence; CE = Conclusion Evidence; LHR = Logical Hypothesis Reliability.
Experiment III: VP = Visual Perception; SL = Spatiotemporal Logic; RA = Rule Application; DD = Domain Depth; CC = CoT Coherence.